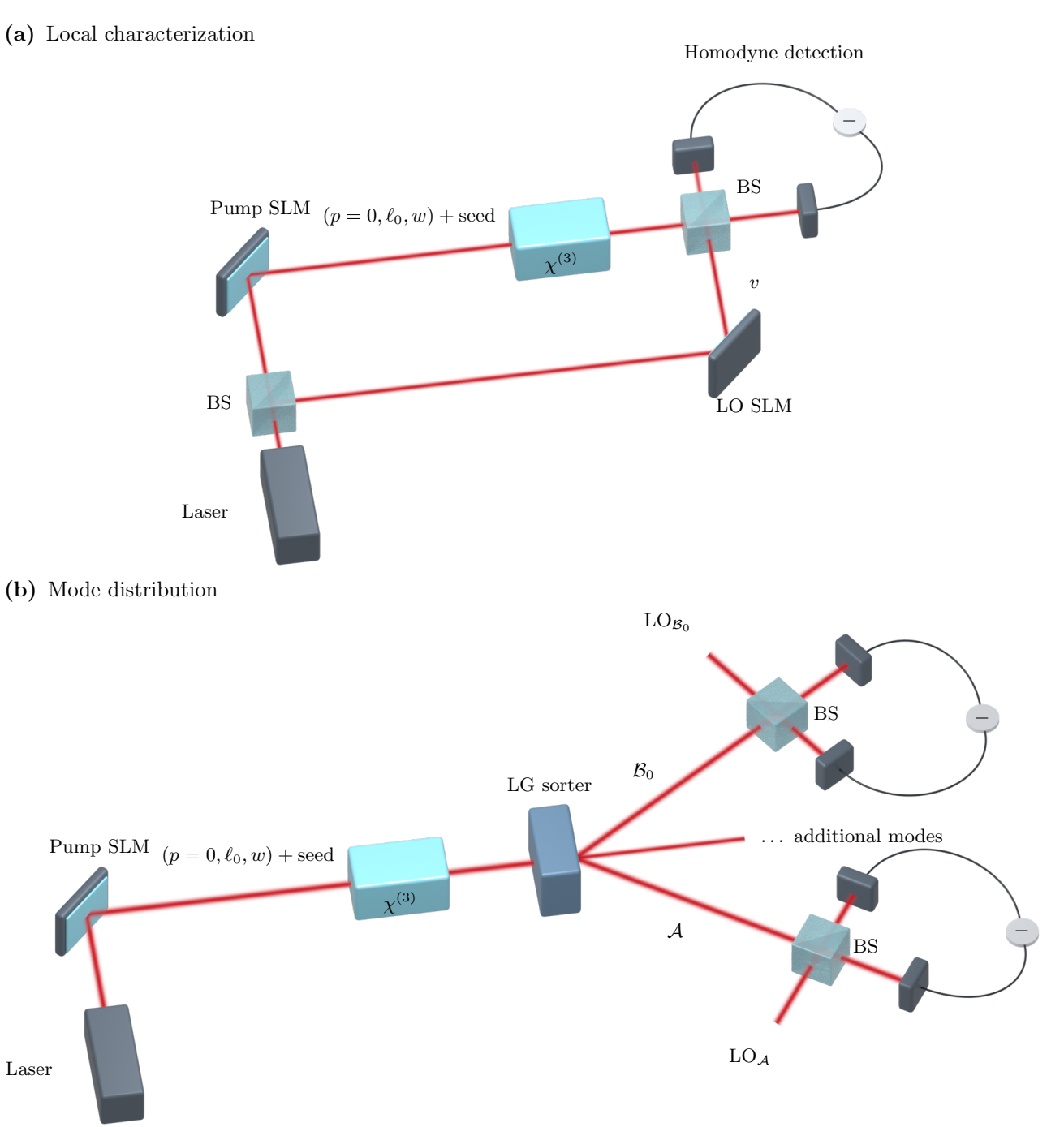

(a) Local characterization
Homodyne detection
BS
Pump SLM
$(p = 0, \ell_0, w)$ + seed
$\chi^{(3)}$
v
BS
LO SLM
Laser
(b) Mode distribution
$\mathrm{LO}_{\mathcal{B}_0}$
BS
$\mathcal{B}_0$
LG sorter
Pump SLM
$(p = 0, \ell_0, w)$ + seed
$\chi^{(3)}$
... additional modes
$\mathcal{A}$
BS
Laser
$\mathrm{LO}_{\mathcal{A}}$